Mean-Variance Optimization

MGMT 675: AI-Assisted Financial Analysis

RICE | BUSINESS
Jones Graduate School of Business

Objective

- The python code that AI will generate to find the tangency portfolio will probably use a different method than you've seen before.
- We'll do a quick overview of the method.

Finding efficient portfolios

The following are equivalent methods:

1. Maximize Sharpe ratio = risk premium / risk
2. Maximize expected return subject to not exceeding a risk limit
3. Minimize risk subject to achieving a target expected return
4. Maximize expected return - penalty parameter times variance

Calculus review

- Suppose $y = x^2 - 2x + 5$ and you want to find the value of x that minimizes y.
- The minimum occurs at the bottom of the curve, where the slope is zero.
  - The slope is the derivative, and the derivative is $dy/dx = 2x - 2$.
  - So, the slope is zero when $2x - 2 = 0$, which implies $x = 1$.
- Bottom line: can solve an equation to find the minimum.

Equations that solve mean-variance problems

- w_i = weight, $\bar{r}_i$ = expected return, r_f = risk-free rate, σ_i^2 = variance, σ_{ik} = covariance.
- 3 asset example: 3 equations in 3 unknowns w_1, w_2, w_3

$$\sigma_1^2 w_1 + \sigma_{12} w_2 + \sigma_{13} w_3 = \bar{r}_1 - r_f$$

$$\sigma_{12} w_1 + \sigma_x^2 w_2 + \sigma_{13} w_3 = \bar{r}_2 - r_f$$

$$\sigma_{12} w_1 + \sigma_{12} w_2 + \sigma_3^2 w_3 = \bar{r}_3 - r_f$$

Matrix multiplication

This is not essential, but it will help to understand code.

- Three equations are represented as

$$\Sigma w = \bar{r} - r_f$$

  where $\Sigma = 3 \times 3$ array of covariances and variances, $w = (w_1, w_2, w_3)$ and $\bar{r} - r_f = (\bar{r}_1 - r_f, \bar{r}_2 - r_f, \bar{r}_3 - r_f)$

- Solution is represented as

$$w = \Sigma^{-1}(\bar{r} - r_f)$$

Tangency and other portfolios

- Given solution (w_1, w_2, w_3) of the equations, divide by sum of w_i to get tangency portfolio
- Given tangency portfolio,
  - Given risk limit (standard deviation), optimal portfolio satisfying the risk limit is

$$\frac{\text{risk limit}}{\text{std dev of tangency portfolio}} \times \text{tangency portfolio.}$$

  - Given target expected return, optimal portfolio achieving the target is

$$\frac{\text{target risk premium}}{\text{risk prem of tangency portfolio}} \times \text{tangency portfolio}$$

Excel Spreadsheet Example

Excel Sharpe Ratio Example (linked on Schedule page)

based on Applied Finance Topic5.1_SharpeRatioExamples.xlsx